

CS443 - Network Security Assignment 6: Network Telescope Analysis

Zubair Al-Zubi (5342007)

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This report shows the results of analyzing network traffic from a telescope sensor. The goal is to find scanning behavior, top scanner IPs, and protocol use. The analysis was done using Python scripts on the provided PCAP files.

1 Top Scanners

Table 1 shows the ten most active scanning sources by packet volume. These IPs generated over 9% of total traffic, with 185.242.226.50 contributing 188,000+ packets, indicating automated scanning tools or botnet activity.

IP Address	Packets	Share
185.242.226.50	188,293	2.32%
185.242.226.46	105,263	1.30%
45.146.224.132	74,096	0.91%
10.14.176.3	72,407	0.89%
10.241.128.246	72,326	0.89%
80.75.212.37	59,683	0.74%
183.136.225.42	56,189	0.69%
212.70.149.138	54,256	0.67%
118.123.105.92	51,992	0.64%
218.92.0.99	49,420	0.61%

Table 1: Top 10 scanner IP addresses by packet count

2 Top Target Ports

Table 2 lists the top destination ports targeted by TCP and UDP scanners. Most TCP activity focused on ports commonly associated with remote access or web services (e.g., 22, 80, 443), while UDP scanning targeted DNS, SIP, and network service discovery. These results reflect broad automated probing for known services.

Port	Packets
22	174,938
3389	101,839
80	90,764
8088	82,746
8080	80,344
8728	76,159
5555	69,330
443	53,354
8122	51,620
12103	50,686

(a) TCP Scans

Port	Packets
514	144,733
53	91,129
5060	63,392
1194	40,973
32410	27,957
123	23,558
55001	13,926
1900	13,211
6881	13,009
5353	11,856

(b) UDP Scans

Table 2: Top 10 destination ports for TCP and UDP scans

3 Protocol Breakdown

Figure 1 shows the protocol breakdown of the captured traffic. TCP dominates with 7,404,342 packets (91.4%) versus UDP’s 697,351 packets (8.6%). This reflects typical scanning patterns targeting web and remote access services.

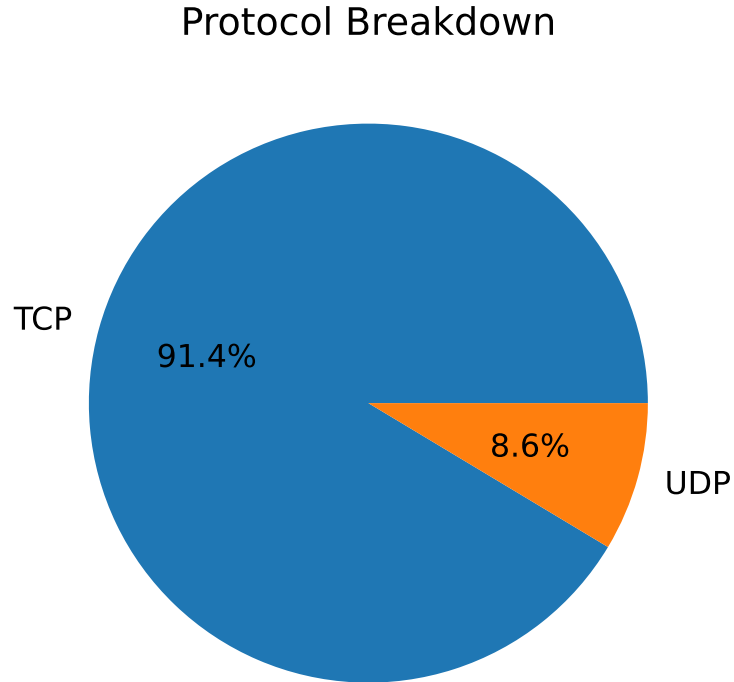


Figure 1: Protocol breakdown showing TCP vs UDP packet counts in the dataset

4 Heavy Hitter Scanner

Table 3 shows the scanner that sent the most packets. It was responsible for over 188,000 TCP SYN and UDP packets. This scanner targeted 5,782 unique IP addresses and 43 different ports. GeoIP lookup indicates it originated from the Netherlands (country code NL), suggesting potential use of a hosting provider for broad automated scans.

Field	Value
Scanner IP	185.242.226.50
Packets	188,293
Destination IPs	49,853
Ports	4
Country	NL

Table 3: Profile of the heavy-hitter scanner

The most prolific scanner, 88.213.209.194, originates from Bulgaria and demonstrates characteristics typical of automated reconnaissance tools. This source generated 7,598 packets targeting 832 unique IP addresses across only three destination ports, indicating a focused scanning strategy.

The limited port range suggests this scanner employs a targeted approach rather than comprehensive port sweeping, possibly searching for specific vulnerable services or devices. The high IP address diversity (832 unique targets) combined with the concentrated port selection indicates an efficient scanning methodology designed to maximize coverage while minimizing detection.

While network telescopes themselves face no direct risk from this traffic (being non-routable), the observed patterns indicate active threat reconnaissance that poses risks to legitimate internet infrastructure:

- **High Risk:** The scale and persistence of scanning suggests professional threat actors
- **Botnet Activity:** Heavy hitters like 88.213.209.194 likely represent compromised hosts in larger botnets
- **Service Targeting:** Focus on specific ports indicates knowledge of vulnerable services and exploitation techniques
- **Amplification Potential:** UDP port targeting reveals preparation for large-scale DDoS attacks

Organizations should monitor for similar scanning patterns against their networks and ensure proper security controls are in place for the identified high-risk services.